Reproducing DoRA for Commonsense Reasoning with Student-Scale Compute

Ian Holloway Younghyun Jung Swathi Selvam Shashwat Modi

Cornell University CS 5782 / CS 4782 Intro to Deep Learning

Motivation & Paper Idea

Full fine-tuning large language models is expensive in memory, storage, and training time. **LoRA** reduces this cost by training low-rank adapter updates while freezing the base model, but the low-rank update alone may limit how pretrained weights can change.[1]

DoRA

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.[3]
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.
- The paper reports improved accuracy over **LoRA** across multiple tasks.

Project goal

- Central question: does **DoRA** improve commonsense reasoning accuracy over a same-scope **LoRA** baseline?
- Reproduce the paper's LLaMA2-7B commonsense reasoning trend.
- Test whether **DoRA** helps most in attention layers, MLP layers, or only when applied broadly.

Methodology

Model, data, and metrics

- Base model: `meta-llama/Llama-2-7b-hf`.
- Training data: `commonsense_15k.json`, a student-scale commonsense subset.
- Evaluation: BoolQ, PIQA, Social IQA, HellaSwag, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA.
- Metrics: per-task accuracy and macro-average accuracy across all eight tasks.

Implementation

- From-scratch PyTorch **LoRA** and **DoRA** adapter layers injected into Hugging Face model modules.
- Frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256.
- 4-bit NF4 Colab runtime presets for T4, L4, and A100 GPUs.

Controlled comparison. Same model, 15k data subset, tasks, prompt/eval pipeline, seed 42, and adapter-only training. Paper-style method settings: **LoRA** r=32, LR=3e-4; rank-halved **DoRA** r=16, LR=2e-4.

Our Independent Experiment: Scope Ablation

Scope	Target modules	Purpose
Full	`q,k,v,up,down` projections	Paper-level reproduction
Attention only	`q,k,v` projections	Test routing/context layers
MLP only	`up,down` projections	Test feed-forward layers

Motivation. This ablation tests whether **DoRA**'s gain comes from attention layers, MLP layers, or coordinated updates across both.

Design. Each scope compares same-scope **LoRA** and **DoRA** adapters under the same evaluation pipeline.

Headline Result

77.41%	79.20%	+1.79	6/1/1
Full **LoRA** macro	Full rank-halved **DoRA** macro	accuracy points	task wins/ties/losses

Figures 1 and 2 show that full-scope, rank-halved **DoRA** reproduced the paper's qualitative improvement over **LoRA** under our student-scale setup.

Macro Accuracy by Adapter Scope

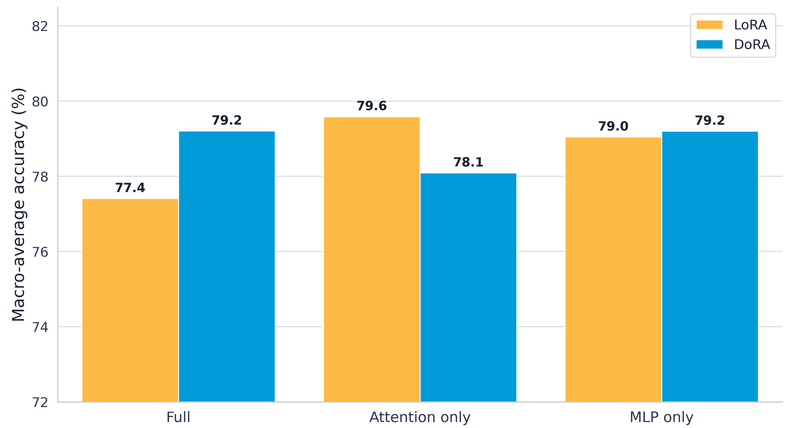

Figure 1. Full-scope, rank-halved **DoRA** improves macro accuracy from 77.4% to 79.2%. Attention-only **DoRA** does not improve over its same-scope **LoRA** baseline.

Scope	LoRA macro	DoRA macro	Delta
Full	77.41%	**79.20%**	**+1.79**
Attention only	**79.59%**	78.09%	-1.50
MLP only	79.05%	**79.20%**	+0.15

HellaSwag: Largest DoRA Gain (+6.0 pts)

Task Format — Grounded Activity Completion

Given a short activity description, select the most plausible four-way continuation. Requires grounded physical-world commonsense.

Context: A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She pours soap on the dog and…

- **(A)** lathers the dog up with a brush before rinsing.
- **(B)** pours more water slowly into the bucket.
- **(C) scrubs the dog down and rinses it thoroughly.**
- **(D)** lets the dog run off and air dry on the lawn.

Full-scope DoRA vs. LoRA. 79.5% vs. 73.5% on HellaSwag — a **+6.0 pt** jump, the largest single-task gain in our study.

Why so large?. Physical activity completion requires tight coordination of attention (routing) and MLP (world facts) layers — exactly the regime where decomposed magnitude+direction updates outperform plain low-rank updates.

Takeaway. **DoRA**'s benefit is not automatic; it peaks when both layer types adapt together.

Official Reference Context

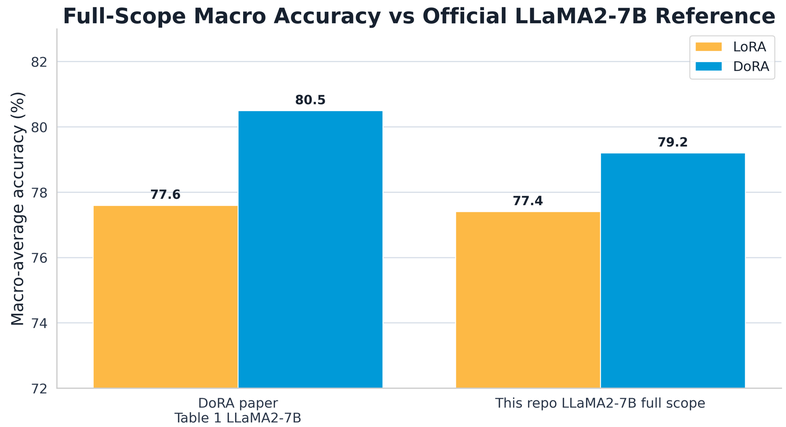

Figure 3. Our result matches the paper's direction: rank-halved **DoRA** improves over **LoRA**, but our 79.2% **DoRA** score remains below the official 80.5% reference.

Interpretation

- The main paper-level claim **partially reproduced**: full-scope **DoRA** beat full-scope **LoRA**.
- Scope ablations show the gain was **not automatic**: attention-only **DoRA** underperformed **LoRA** on 7/8 tasks; MLP-only **DoRA** was roughly tied.
- Key takeaway: **DoRA** appears most useful when attention and MLP transformations adapt together, not as a replacement for one module group.
- Key limitations: 15k subset instead of full data, seed 42 only, quantized Colab runs, and a local implementation rather than the exact official recipe.

Conclusion & Future Work

Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points**.
- The reproduction supports the qualitative value of weight-decomposed low-rank adaptation.
- The ablation adds a caution: applying **DoRA** narrowly can remove the benefit.

Next steps

- Train on the full `commonsense_170k.json` dataset.
- Repeat all six conditions across multiple random seeds.
- Add parameter-count, GPU-memory, and wall-clock cost comparisons.

References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.

[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.

[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.